

Artificial Intelligence in Breast Cancer Diagnosis: “Synergy-Net” in Campania FESR-POR (European Fund of Regional Development—Regional Operative Program) Research Project



Domenico Parmeggiani, Giancarlo Moccia, Pasquale Luongo, Francesco Miele, Alfredo Allaria, Francesco Torelli, Stefano Marrone, Michela Gravina, Carlo Sansone, Ruggiero Bollino, Roberto Ruggiero, Paola Bassi, Antonella Sciarra, Simona Parisi, Francesca Fisone, Maddalena Claudia Donnarumma, Chiara Colonnese, Paola Della Monica, Marina Di Domenico, Ludovico Docimo, and Massimo Agresti

Abstract *Background:* Nowadays, mammography and DCE-MRI are the gold standard for breast cancer screening. Scientific experience demonstrate an increasing application of Echo elastosonography in breast cancer diagnosis, for this reason, within the Synergy-Net project the objective is to create a system based on machine

D. Parmeggiani (✉) · G. Moccia · P. Luongo · F. Miele · A. Allaria · F. Torelli · P. Bassi · A. Sciarra · S. Parisi · M. C. Donnarumma · C. Colonnese · M. Agresti
Department of Advanced Medical and Surgical Sciences, University of Campania “Luigi Vanvitelli”, Naples, Italy
e-mail: domenico.parmeggiani@unicampania.it

S. Marrone · L. Docimo
Department of Woman, Child and General and Specialized Surgery, University of Campania “Luigi Vanvitelli”, Naples, Italy

M. Gravina · C. Sansone
Department of Electrical Engineering and Information Technology (DIETI), University “Federico II”, Naples, Italy

C. Sansone
CINI, ITEM Laboratory “C.Savy”, Via Cintia 21, Naples, Italy

R. Bollino · R. Ruggiero
Bollino.It SpA, Via Delle Ustrie 31, Naples, Italy

R. Ruggiero · S. Parisi · F. Fisone
Division of General, Mini-Invasive and Obesity Surgery, University of Campania “Luigi Vanvitelli”, 80131 OncologicalNaples, Italy

P. D. Monica · M. Di Domenico
Department of Precision Medicine, University of Campania “Luigi Vanvitelli”, Naples, Italy

learning algorithms, a CAD developed with CNNs, capable of representing a decision support in the analysis of echo-elastic sonographic images. *Results*: 315 female patients at the “Vanvitelli” Breast Unit were subjected to digital 3D mammography tomosynthesis and advanced ultrasound (quantitative Echo elastosonography) and comparative analysis of the two methods. We collected 70 breast cancer, 200 benign pathologies (dropout of 11.1%) and we compared two different homogeneous by composition subgroups (35 with cancer and 100 with benign pathology 20.5%), relating to the diagnostic predictive capabilities of 3D digital mammography and quantitative elastosonography: Group A age 40–50 years, Group B age 50–60 years. Results demonstrate fair predictive performances of echoelastosonography versus traditional mammography, especially in the first group (40–50 years), $p < 0.05$ in favor of elastosonography regarding sensitivity and accuracy; performances that appear substantially comparable in the second group (50–60 years). We also tested the diagnostic predictive capabilities of the CAD Synergy-Breast-Net achieving encouraging results: Sensitivity 80%, Specificity 72%, Accuracy 74%, Negative Predictive Value 84.7% and Positive Predictive Value 83.3%. *Conclusion*: From this perspective, the screening of patients aged < 40 years and integrate the main screening activities in patients aged between 40 and 50 years integrated with increasingly high-performance machine learning systems, could represent a valid alternative in the future.

Keywords Breast cancer · Elastography · FibroScan · CNNs · Artificial intelligence

1 Introduction

This study is part of a ReS project (Synergy-Net: Research and Digital Solutions in the fight against oncological diseases). Its objective is the development of a Synergy-Net technological platform aimed at improving and strengthening oncological cancer prevention. In this study, we refer exclusively to experimental developments produced in the diagnosis of breast cancer.

Breast cancer is the most common neoplasm in women. In 2019, more than 50,000 women fell ill with this neoplasm in Italy. The reduction in the average age of onset of this neoplasm is also certain. This data affects screening diagnostics in two different ways: The first is that young women are not enrolled in screening programs and the second is that in young breasts clinical and instrumental diagnostics are more difficult due to the density of the breast. The reduction in sensitivity of mammography in dense breasts (classes C and D of mammographic density) is certain. Furthermore, if we consider that early diagnosis is an imperative to achieve patient recovery, we understand how breast diseases constitute a challenge for diagnostics. Added to this is the expansion of the audience of women to be called for screening also to the 40–50 years age group, with risk factors, in which the possibility of dense breasts is significantly greater. Hence, increasingly refined technologies in the study of the

dense breast, such as mammography with 3D digital tomosynthesis and mammography with contrast medium. The need therefore emerges that, in screening programs, reference cannot be made to mammography alone but that a complex strategy is necessary that can lead to personalized diagnostic paths based on different factors and also using different diagnostic techniques. Hence, increasingly refined technologies in the study of the dense breast, such as mammography with 3D digital tomosynthesis and mammography with contrast medium. The need therefore emerges that, in screening programs, reference cannot be made to mammography alone but that a complex strategy is necessary that can lead to personalized diagnostic paths based on different factors and also using different diagnostic techniques. Hence, increasingly refined technologies in the study of the dense breast, such as mammography with 3D digital tomosynthesis and mammography with contrast medium. The need therefore emerges that, in screening programs, reference cannot be made to mammography alone but that a complex strategy is necessary that can lead to personalized diagnostic paths based on different factors and also using different diagnostic techniques [1, 2].

It therefore appears necessary to plan on the basis of various parameters which can lead to a rationalization and, at the same time, a reduction in healthcare spending from investigations that are not useful for diagnosis purposes [3].

For this reason, we have tried to identify a diagnostic path that is not only instrumental but which can be “intelligent”, that is, integrated with the most modern machine learning technologies, capable of helping healthcare workers in choosing a diagnostic path-personalized therapy for each individual woman.

The primary objective is to reduce spending on unnecessary investigations as well as optimize knowledge in the field of breast cancer diseases.

Nowadays, mammography and contrast-enhanced magnetic resonance imaging (DCE-MRI) are the gold standard for breast cancer screening. In particular, DCE-MRI, digital mammography with 3D tomosynthesis and contrast-enhanced mammography are second-level diagnostic tests, used above all for their ability to capture physiological aspects of the tissues under analysis. Unfortunately, compared to mammography, they are very expensive and, its long acquisition time (RNM), irradiation (3D tomosynthesis) and the level of expertise required (contrast mammography) place limits on the number of patients who, every day, can be analyzed and therefore to their applicability in the screening phase. For this reason, within the Synergy-Net project the objective is to create an AI system that can allow the use of further technology, i.e., a CAD developed with CNNs, for the analysis of echo elastosonographic images, a rapid ultrasound method, painless and relatively inexpensive. This technology integrated into an online platform and with the progress of increasingly advanced and economical ultrasound technologies, could in the future represent the elective procedure in the screening of patients aged < 40 years and integrate the main screening activities in patients aged between 40 and 50 years. Painless and relatively inexpensive. This technology integrated into an online platform and with the progress of increasingly advanced and economical ultrasound technologies, could in the future represent the elective procedure in the screening of patients aged < 40 years and integrate the main screening activities in patients aged between 40 and 50 years. Painless and relatively inexpensive. This technology integrated into

an online platform and with the progress of increasingly advanced and economical ultrasound technologies, could in the future represent the elective procedure in the screening of patients aged < 40 years and integrate the main screening activities in patients aged between 40 and 50 years [4–10].

2 Materials and Methods

In a first phase, we evaluated on the basis of personal experience and literature data what anamnestic and instrumental information can lead to a more or less complex diagnostic path. We took into consideration several easily identifiable factors such as clinical criteria and the calculation of risk factors (for example with the Gail method) with particular emphasis on age, family history, nulliparity, age of the first pregnancy, previous surgery for cancer breast or for potentially evolving anomalies, hormonal therapies (stimulation) (Table 1).

Below we have taken into consideration specific data from traditional mammography with findings such as density in relation to age the detection of newly appearing mammographic opacities also in relation to age the type of micro-calcifications detected and their variation in number and extension in two subsequent tests the type of opacity detected in relation to age, the volumetric increase in the opacities in subsequent checks and, both clinical and mammographic findings, the alterations of the profile and skin edema (Tables 2, 3, and 4).

Based on our experience, we have given a multiplicative value in order to help engineers standardize and quantify a good part of the information provided to make

Table 1 Clinical criteria

	40–50 years	51–60 years
Profile alterations	X 1.5	X 2.5
Unilateral nipple retraction as a new symptom	X 1.5	X 2.5
Unilateral mono- or oligo-porous blood or serous secretion	X 1.5	X 2.0
Megalic axillary lymph nodes	X 1.5	X 2.0

Table 2 Mammographic density-classes C and D

Age	Value
40–50 years	1.5
51–60 years	2.5

Table 3 Newly appearing mammographic opacity compared to previous exams

Age	Value
40–50 years	2
51–60 years	3

Table 4
Micro-calcifications:
typology and variation in
number and extent

	Value
Not suspicious	1
Suspicious	2.5
Suspicious inscribed in irregular opacity	4
Increase in number and extension	4

an evaluation that is as reliable as possible, independent of individual interpretative variations. Once the data to be recovered and integrated with the images had been defined, the current limits of the screening protocols in use were defined, thus highlighting the diagnostic limitations in juvenile dense breasts. The multiplicative values are relative to:

We have therefore focused our attention on diagnostic images acquired with a latest generation ultrasound system: Supersonic Mach 30, in particular, is an ultrasound supported by a processor that works with Ultra-Fast technology, which has 5 times more computing power. Intelligent signal processing uses algorithms in image reconstruction, based on neural networks, capable of acquiring up to 20,000 images per second, furthermore TriVu imaging combines real-time simultaneous imaging of B-mode, i ShearWave™ PLUS elastography and Angio PLUS imaging, allowing doctors to view anatomy, tissue stiffness and blood flow on the same image simultaneously. This mode provides access to vascularity-guided measurement and tissue stiffness. This real-time technique is able to measure tissue stiffness in a non-invasive way, but above all it is able to provide a quantitative evaluation in Kilopascals (kPa) of tissue hardness, a fundamental element for limiting subjective variables, a crucial methodological bias in training of an ultrasound cad. Once the diagnostic protocols and the type of data to be acquired were defined, we began to work on the protocol and machine learning algorithms to apply to our study for the creation of a CAD developed with convolutional neural networks. We enrolled a group of 315 female patients aged between 35 and 72 at the Department of Advanced Medical and Surgical Sciences and the Breast Unit of the University of Campania “Luigi Vanvitelli”, from January 2021 until December 2021. Years, average age 55.4 years, subjected to digital mammography with 3D tomosynthesis and advanced ultrasound investigation (Echo with quantitative elastosonography) and we compared the predictive capabilities of the two methods: The patients were enrolled and subjected to Micro-histological Biopsy, RNM, staging investigations and subsequent GOM protocol with related indications for neoadjuvant therapy and/or surgical therapy. In the sample under examination we had a dropout of 11.1% due to lack of acquisition standardization methodology (some of the most common echographic dataset weaknesses) and dataset incompleteness (some of the most common CNNs weaknesses); we analyzed two different subgroups made homogeneous by composition (35 with cancer and 100 with benign pathology 20.5%), in relation to the diagnostic predictive capabilities of 3D digital mammography and quantitative elastosonography [11].

- Age 40–50 years.
- Age 50–60 years.

The technological platform created for Synergy-Net has acquired a copious dataset of clinical-anamnestic information and over 280 ultrasound scans acquired in Dicom format, with a first image clean of comments, indications and/or measurements, a second with reference to the site, in at least two dimensions, with perimetric reconstruction of the Region of Interest (ROI) and volumetric analysis of the lesion, a third image with geometric area (rectangle) including the lesion, analysis of the vascular microcirculation and in the same image contemporary analysis of the elastographic index of the lesion in kPa, with chromatic representation of the lesions (represented in a chromatic scale that goes in order from the hardest tissue to the softest, from the chromatic shades of red to the color blue). The images were saved with a patient identification number and a classification code relating to the USBirads classification provided by the operator (e.g., K1USbirads5eco or ecoelastovasc, etc.). After selecting the ROI from the echo-elastosonographies, a CNN-based system is used for classification.

In particular, AlexNet was used to distinguish nodules between different classes. To handle the small size of the dataset, the network involved is pre-trained on the ImageNet dataset and fine tuning is used to adapt the network to the specific task to be solved. In order to increase the size of the dataset, 5 square patches are extracted from the ROI highlighted during the elastosonography. The final prediction is carried out considering a vote among the labels relating to the extracted patches, weighted based on the probability of prediction.

The CAD-CNNs Synergy-Breast-Net created was finally tested in the field during presentation days of the Project at a local level in a high incidence area, on 190 ultrasound images of approximately 100 pre-selected patients, of which 35 presented highly suspicious nodules, all of which were then confirmed by micro-histology as malignant.

This sample of 35 patients with malignant neoplasms was subjected to a complete analysis of the entire genome, approximately 46,000 variables/patient, which enriched the amount of data of the expert system developed and for which, at the moment, no data is yet available relevant for potential correlations.

- The first objective of the study is to verify in patients, based on the two different age classes, the predictive performance of echo-elastography compared to digital mammography.
- The second objective is therefore to confirm, at least preliminarily, the predictive performances of the CAD-CNNs Synergy-Breast-Net.

3 Results

Performance is evaluated in terms of Accuracy (ACC), Specificity (SPE) and Sensitivity (SENS). Positive Predictive Value (PPV) and Negative Predictive Value (NPV) and in relation to the first objective the results certainly appear interesting, demonstrating fair predictive performances of ultrasound elastosonography vs traditional mammography, especially in the first group (40–50 years), with statistically significant differences ($p < 0.05$) in favor of elastosonography regarding sensitivity and accuracy; performances that appear substantially comparable (statistically insignificant differences) in the second group (50–60 years) an interesting predictive element in elastosonography is the presence of nodular elastosonographic values above 100 kPa (Table 5).

Regarding the second objective, the result, although largely conditioned by the numerical limitation of the sample, appears promising.

The performances of the CAD Synergy-Net are all statistically significantly worse than both the mammographic and elastosonographic performances (if a statistically significant difference is excluded not significant in relation to sensitivity compared to digital mammography in group I) (Table 5). The non-exceptional performances largely depend on the small size of the available dataset, but can certainly be improved by integration with the clinical-anamnestic dataset, by numerical implementation and the consequent completion of the learning curve. Integration with gene profiling of cancer patients could also help reveal new development and research scenarios in the future (Table 6).

Table 5 Diagnostic performance between mammography and elastosonography based on age

Study	Mammography 40–50 years	Eco-Elasto 40–50 years	Mammography 51–60 years	Eco-Elasto 51–60 years
Sensitivity	28/35 (80.0%)	32/35 (91.4%)	30/35 (85.7%)	31/35 (88.5%)
Specificity	91/100 (91.0%)	95/100 (95.0%)	96/100 (96.0%)	97/100 (97.0%)
Accuracy	119/135 (88.1%)	127/135 (94.0%)	126/135 (93.3%)	128/135 (94.8%)
NPV	100/107 (93.4%)	100/102 (98.0%)	100/105 (95.2%)	100/103 (97.0%)
PPV	35/37 (94.5%)	35/36 (97.2%)	35/36 (96.9%)	35/36 (97.2%)

Table 6 Ultrasound diagnostic performance of the CAD Synergy-Net

SENS	SPE	ACC	NPV	PPV
28/35 (80.0%)	72/100 (72.0%)	100/135 (74.0%)	100/118 (84.7%)	35/42 (83.3%)

In compliance with the architectural document, a python script was also created that performs the segmentation of neoplastic lesions in elastosonography. This script represents the artificial intelligence module used by the Synergy-Net application for the analysis of images relating to breast lesions. When a more performing classification model is available, it will be sufficient to replace just one file, making the update operation totally transparent to the CAD application.

4 Discussion

Both quantitative elastosonographies with ShearWave and integrated diagnostics with machine learning systems based on CNNs represent some of the most innovative technologies in the diagnosis of breast lesions, as demonstrated by copious literature on the subject, thanks to the real-time mapping of the stiffness of the breasts. Breast lesions, an element that provides further diagnostic information useful for improving patient management. Elastographic imaging constitutes a complementary tool for the management of breast cancer patients for:

- Diagnosis and characterization of breast lesions [12].
- Biopsy planning and treatment [13].
- Therapy planning and monitoring [14].

ShearWave™ PLUS elastography can contribute to the diagnostic phase of breast lesions and therefore have a positive impact on patient management¹. It can also help locate lesions during ultrasound-guided biopsy [15].

It is clear that we are talking about high-end and latest generation ultrasound machines, certainly not ideal for a screening protocol, but the objective of our study seeks to acquire the greatest amount of ultrasound information, in order to imagine, in the future, an online, open source tool capable of interfacing with the largest number of ultrasound images and supporting diagnosis with a CAD based on convolutional neural networks. There are numerous and constantly evolving screening programs proposed with a view to an early diagnosis of breast cancer, however, over the years and with the development of new diagnostic techniques, the awareness that mammography, alone, is an insufficient tool. The integration of mammographic diagnostics with heavier or contrast imaging techniques or with ultrasound examination is now increasingly widespread, in particular by enhancing the interpretation of imaging with technologies based on Machine Learning systems [16–20].

However, not all diagnostic units have kept pace, even less with the most advanced diagnostic techniques such as magnetic resonance imaging and mammography with contrast medium.

It is clear that equipping every diagnostic unit in the area with such equipment is absolutely impractical both due to the costs of equipment and specialized human material.

The industry is developing complex screening mammogram analysis systems that can help in two ways:

- By analyzing mammograms and directing to a second level only those that the system deems worthy of human control and subsequent investigations.
- Electronically analyze mammographic images deemed uncertain or suspicious.

While waiting to have expert systems capable of detecting uncertain or suspicious mammographic images, the need to minimize mammographic errors which often entail unnecessary additional costs, reducing the possibility of error to a minimum, appears clear. From this perspective, the latest generation ultrasound investigations, integrated with increasingly high-performance machine learning systems, could represent a valid alternative in the future, especially in younger patients.

5 Conclusion

This work described the Synergy-Net, an ongoing project aimed at creating a technological platform to support early oncological diagnosis based on the integration of interoperable communication and also a clinical data management system that exploits AI. Due to its deeply interdisciplinary nature, the Synergy-Net system was designed as a modular CAD where each module cooperates with the others, under the guidance of an orchestrator, to provide the required computation. This interdisciplinary nature has allowed us to work in parallel on different organs, exploiting common architectures, solutions and ideas and designing specific solutions for each patient.

The downside is that the project’s progress is not aligned, mainly due to different needs (e.g., conditions that must be met to engage a patient). This has been further complicated by the ongoing COVID-19 emergency. The result is that, while the design of the artificial intelligence algorithm for some organs has been completed (or almost completed), for other organs we are still in the data collection phase. Despite this, while working to complete ongoing activities, future steps are already being planned. There are two main ideas that the project will work on in the near future.

- The first idea is to exploit modern ultrasound elastosonography as a low-cost methodology with no side effects for the early diagnosis of oncological diseases for all tumor types considered in the Synergy-Net project and in which it is applicable.
- The second idea is to provide integrated prevention through data via fusion techniques. The objective is to simultaneously analyze information from multiple sources, in order to provide an integrated tool built on the medical knowledge of each specialist.

This will be further improved by the use of DNA sequencing tests aimed at finding correlations between mutations and imaging abnormalities that lead to cancer.

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Declaration

- Ethics approval and consent to participate

This study followed the ethical principles of the Declaration of Helsinki and every participant was signing and approving the consent by the Institutional Review Board at “Policlinico” Hospital of the University of Campania “Luigi Vanvitelli” in Naples, Italy. Participation in the study was voluntary. Before inclusion in the study, study staff explained the purpose of the study and informed consent form was secured from each participant.

- Consent for publication

We have had authorization for publication like our Hospital Privacy Policy already require.

- Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

- Competing interests

The authors declare that they have no competing interests.

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